MKTG/STAT 4760/7760
Spring 2025

Project 1 Guidelines and Advice

Key Elements

Task	Implement suitable NBD models on one (or, preferably, several) count datasets on a topic of interest to you.
Deliverable	A well-written report detailing your findings. The paper should read as a narrative about your data, objective, and findings – not a technical report with a bunch of Excel screenshots in it.
Length	2500 words (not strict but no one wants to read a long paper)
Audience	Your classmates, who are already familiar with the basics of the NBD
Grading	WHOOPPEE – based on classmates' ranks of your paper compared to others. You cannot include ANY personally identifiable information in the paper.
Due Date	Wednesday February 25, 9:00 PM EST – hard deadline!

Identifying topics/datasets

You really want to start looking for data ASAP; the right dataset can make or break your project, so we will keep nudging you to ensure that you find good data (and do a good job on the paper).

Here are some quick descriptions of past projects. We're not suggesting that anyone repeat these specific analyses (in fact, we discourage it). Still, we hope these examples inspire you to think of some other new and interesting applications for count models:

- *Marketing behavior* (e.g., renting streaming movies, number of portable electronic devices owned, number of leads generated by salespeople)
- *Other business topics* (e.g., frequency of manufacturing defects, number of subsidiaries/business units under a corporation, number of job-search sites used)
- *Sports* (e.g., top ten finishes by pro golfers, NBA players fouling out of games, yellow cards received in soccer, number of runs by cricket batsmen)
- *Science* (e.g., frequency of tornadoes within each state, number of thunderstorms, earthquakes, etc.)
- *Other behaviors* (e.g., traffic accidents at various intersections, frequency of use of EZPass, number of houses lived in, frequency of illegal drug use)

You should be able to come up with many possible examples within these broad groups or outside of them. We don't care about the domain as long as it's interesting, the data are appropriate, and the analysis is well done. Ideally, you'll examine several related datasets for a given topic (e.g., data for different regions, customer segments, etc.) and offer some managerially appropriate comparisons across them, as well as any words of caution about the data (e.g., its representativeness of the broader population) and your conclusions (e.g., their generalizability to other contexts).

We highly recommend that you check out the abundance of resources that the wonderful folks at Lippincott Library have assembled specifically for this course; see the "Resources for datasets" link on the "Project 1

stuff" page. You can also use that link to schedule a consultation with a librarian there – they are happy to chat with you. You should do some exploratory investigations first, then reach out to help refine your search.

Of course, there are many other data archives available on the internet. Feel free to seek out some sources and discuss them on Ed Discussion (without getting too specific about your own project idea). Or, if you prefer, please bring them to our attention privately.

But beyond these suggestions and your own creativity/resourcefulness, do not ask the teaching team for directions or feedback regarding datasets. We're not mean or lazy, but we want you to figure this out on your own. Identifying a suitable dataset (and figuring out how to wrangle it for the project) is part of what we're assessing. All the more reason to get going early in your hunt for a good dataset.

The analysis

The assignment is straightforward: find a rich count dataset (or, better yet, a set of related datasets) and fit appropriate count models. It really should be a setting that calls for a traditional count (i.e., NBD) model. Discrete duration models (from Session 1) and choice models (to be covered in Session 4) might be OK if there's something you're very passionate about, but that will make it harder for your evaluators to compare your paper to others. So we strongly encourage you to stick with standard count data (and NBD models).

This might seem like a silly point, but make sure you use a ***count*** dataset. It's not right to take continuous data and chop it up into discrete buckets just so that you can jam it into a histogram. You must be able to tell an individual-level story of a discrete count process. Part of the paper's evaluation is whether the model you choose is appropriate for your data. So, if you try to fit an NBD model to data that aren't really counts, that would be bad; it would indicate a lack of understanding of when to use these models. There are few things worse in quantitative modeling (or home repairs, for that matter) than using the wrong tool for the job.

Here's another way to think about it: ideally, your dataset should be a count summary of a process that occurs repeatedly over time (e.g., seeing billboards, buying coffee creamer, Facebook sessions), and occurs at different rates for the individual-level observation units (e.g., customers) of interest.

One subtle but common exception to this advice is a "choice" dataset that has an extremely high (and basically unreachable) upper bound (e.g., number of days in a year with a tornado in a particular city). (We'll cover these models starting in Session 4.) But even then, one should at least acknowledge the fact that it is, in truth, a choice process, not really an unbounded count one. There is no reason to run any beta-binomial models (and if your dataset really demands one, then it might be a bad option for this assignment).

Also, make sure that these repeated occurrences aren't too frequent: we want count data that are relatively sparse, i.e., similar to the datasets we've used in class and homework. If the counts are too large then you don't really need a probabilistic model at all – the observed means will tell you everything about the underlying propensities, etc.

Keep in mind that your individuals might have different values of "t," i.e., different lengths of observation windows. If so, you should account for those differences, as discussed in class. Keep in mind that when you enter "Varying T world" lots of things get weird; it's much harder to rely on simple rules of thumb, such as those that we discussed in Session 2A. Here's another: in class we saw that adding a spike to a cell will make that cell perfect, i.e., actual=expected. But that's not necessarily the case in Varying T world. Of course the cells should be close with the extra parameter there to balance things, but it won't be an exact match anymore. So you have to proceed carefully and maybe consider a wider range of spikes and other model fixes than you would in Simple Histogram world.

It's very desirable to run and compare separate models for subgroups of your sample, e.g., one model for males and another for females. Make sure you carefully compare these sub-models (e.g., interpret the different values of their shape parameters) and test for differences across them.

But be careful about running models across different periods of time. It's perfectly fine to compare parameters across different time periods for the same individuals, with three caveats:

a. You can't run an LRT – it is only valid when you're comparing different cross-sectional subsets, not different time periods for the same individuals. You could (and often will) get a negative LRT in this case.
b. While you can try to explain the differences over time qualitatively, don't try to model them. We'll start doing this in a few weeks.
c. Make sure that both periods contain the exact same set of individuals; if some are entering/leaving the sample over time, then the parameters from one time period to another are much harder to compare.

Beyond the model specification, think carefully about what kinds of inferences you can make, and what they tell us about the phenomenon of interest.

Whatever dataset(s) you choose, we want you to demonstrate mastery over the methods we have discussed in class and show it off with excellent writing skills.

The write-up

As you prepare your paper, you should keep in mind that your target audience consists of your classmates, who are highly familiar with all of the basic concepts we have discussed in the course. Thus, it is unnecessary to be overly formal/technical, e.g., don't include any formulas unless they're essential. Likewise, there is no need for any review/reintroduction to the methods used, or MBA buzzwords, or managerial claims that go beyond the scope of your dataset and model(s). At the same time, don't be too silly/informal/colloquial. Your evaluators will take this analysis/writing task seriously, so you should also do so.

With this high degree of seriousness in mind, we strongly recommend that you familiarize yourself with the WHOOPPEE ("Wharton Online Ordinal Peer Performance Evaluation Engine") platform before you complete your paper. Please review the links on the WHOOPPEE page (available from our Canvas homepage) and post any questions about it on Ed Discussion.

Very important: to ensure fairness in grading, do NOT put your name, your Penn ID, or any other personally identifying information anywhere on your paper. WHOOPPEE will check for PII and reject papers that seem to have any. Be careful and double-check this before submitting your paper.

The paper should be no longer than 2500 words. This isn't a formal restriction, but remember that your classmates will only have so much interest in any topic. There is no need for extensive introductions, literature reviews, or anything like that. Just tell the reader about the interesting aspects of your dataset/analyses.

Not only should your paper do a good job summarizing the results of your work (as well as the implications, next steps, etc.), but you can briefly discuss the process that led you there. That doesn't mean you should write the paper entirely in a chronological "I tried this, then I tried that" format. A little of that is fine, but you must decide which models deserve the most attention (and which should be ignored completely).

You should candidly acknowledge the shortcomings of your analysis, your dataset, and (perhaps) your own capabilities in carrying out the project. Since we have not yet discussed methods to deal with non-stationarity or covariates, it would be a mistake to try to fully capture such stories for this first paper. But if you see evidence

of such departures from the "plain vanilla" NBD model, don't hesitate to point them out and speculate about the possible causes and remedies.

Overall, we want a beautifully written paper that is self-sufficient – all relevant analyses should be incorporated and appropriately documented. We really care about the quality of the writing and the document's formatting, so don't cut corners in this regard. Correct grammar and spelling matter – proofread carefully!

Besides excellent writing, we expect to see good use of figures/graphs to help tell your stories most effectively. Likewise, the paper should be nicely organized/formatted. Overall, you should be proud of what you submit and use similarly high standards when it comes time to judge your batch of papers to evaluate through WHOOPPEE.

Finally, make sure that the paper is self-contained – the relevant parts of any spreadsheets you refer to should be included in the document and clear to the reader. While we require you to submit your full spreadsheets, you should not expect your readers to examine them.

Lots more specific details

The points below were collected from various other documents I shared before/after Project 1 in past years. Some are redundant with the guidance above and some might be irrelevant to your particular analysis. But we hope they're helpful to think about as you proceed with the project.

1. A good executive summary is helpful.

2. Provide careful (but concise) background/motivation for the question(s) you are addressing. Provide complete information about the origins of the data – if they're from a website, give the URL; if they're from a publication, include a copy of relevant tables/figures in the paper (perhaps in an appendix) and upload the complete paper to Canvas).

3. Spend a bit of time talking about the nature of the data and possible concerns associated with it. It's nice to include a sample of the raw data, a histogram, and appropriate summary statistics.

4. You must have a set of distinct individuals (e.g., distinct customers, teams, countries) that provide the source for heterogeneity. Different instances for the same individuals (e.g., multiple games for a given team) do not provide a natural source of heterogeneity – each is just a different "spin of the Poisson wheel." Likewise, using different years as an "individual unit" to model count data aggregated over different individuals (e.g., customers) does not make a lot of sense.

5. There's no benefit in organizing your paper explicitly around the "seven steps." This is a pedagogical rubric to help you understand the model-building process, but it's not necessarily a good framework for a paper.

6. Don't make your paper a chronological "journal" of all the models you ran. It's not necessarily a tragic error to do this, but a more thoughtful write-up is generally more effective.

7. Count data implies meaningful integer values. Grouping continuous data into bins does not create count data; you can't use Poisson-based models in such instances. Similarly, if the counts are very large, it's probably a bad dataset for this project.

8. Make sure you understand the difference between counting and choice data. If there are a finite number of count opportunities, it's really a choice model. If no individuals get anywhere close to the upper bound then it's fine to run a count model, but you should note that it's not exactly a count process.

9. And in such instances (as well as other count settings), it's very important to account for the fact that different individuals may have a different number of opportunities. Each person would have a different "T" in the Poisson or NBD model (or a different "N" for the beta-binomial).

10. You should also be clear about the difference between counting and timing data. While timing data may be expressed in discrete units (as in Session 1), it's hard to conceptualize it as arising from a Poisson count process.

11. If appropriate, provide a quick overview of the models you plan to run. It's great to see hunches/hypotheses about which model will work best under different circumstances. Likewise, it's nice to see guesses about the parameter values you expect to uncover.

12. It's helpful to provide a table summarizing your models' key characteristics (e.g., parameter estimates, LL, chi-square test).

13. There's no reason to run plain Poisson models, etc., if you know they're not going to work well. Stick with reasonably plausible models.

14. Be very clear about the use (and justification) of shifted and/or left-truncated models.

15. I really wanted students to look carefully at the parameter estimates and:
    a. Make sure they're logically consistent with the data and other parameters reported in the paper (e.g., look at the means)
    b. Explain what they imply (e.g., interpretation for r)
    c. Consider doing something with the results (compare parameters across datasets, extend time horizon, examine concentrations, etc.)
    d. And this generally <u>doesn't</u> include doing things with the means. You don't need a model to compare means. It's fine to compare them, but that's generally not a benefit that arises from the model.

16. Don't overdo it with the chi-square goodness-of-fit test. It's often a useful diagnostic, but you shouldn't have blind faith in it (for instance, you still need to look at the histogram). And when the test is highly significant (which happens automatically for a large dataset), you should offer some specific descriptive reasons why that's the case. And make sure you heed the guideline about E[x]>5 for most (if not all) of the cells.

17. Remember that all chi-square p-values >.20 are basically indistinguishable – they all indicate the lack of a meaningful deviation, so don't compare high p-values to each other. Ditto for any p<.001.

18. Furthermore, just because you may need to bin the sparse data in the tail in order to run the chi-square test, that doesn't mean that you should run the model on the binned data. Don't let the chi-square test influence the way you organize the data or run the models. Do all the required binning afterwards (unless you have some reason not to trust the granular data in the tail, e.g., survey data where people report too many round numbers).

19. Be careful when calculating the number of degrees of freedom. It's a very common mistake to ignore this when adding another parameter to a model.

20. We generally avoid using MAPE for histograms, since all the %'s must add to 1. It's not necessarily wrong to use it, but it's not quite as meaningful as it is for the holdout sample of a longitudinal model. (You've never seen me do this, so why start now?)

21. Adding a spike can never make a model fit worse. At worst, the spike won't be significant. Make extra super double sure that you believe in (and convey) the story behind any spike you include in the model.

22. Try to avoid the use of a spike for a right-truncated histogram bin (e.g., for a 10+ cell) – that's generally a bad thing to do (and again, you've never seen me do this).

23. It's quite risky to make up new measures or use techniques that we haven't covered in the course.

24. I'm often surprised (and thus a bit disappointed) to see how few people tried multiple estimation methods (e.g., M&Z), in addition to MLE. Not a big deal, but I do want you to keep these other methods in mind – then can be very useful at times.

25. It's great to examine the model across multiple datasets, but you have to look beyond LRT p-values. Are there meaningful patterns in the parameters? Be clear and thorough in describing differences across the separate sub-models. Also just comparing just observed means is not enough: we can get those (and other summary statistics) without building a model at all. It is more interesting/useful to look at the underlying heterogeneity distributions and their differences across datasets.

26. Make sure your exhibits and appendices are well-formatted. Label them carefully and make specific reference to them in text. If you don't refer to a particular exhibit, then don't include it in the paper.

27. Don't refer to anything in the actual spreadsheet (that isn't included in the paper/appendix).

28. Point out relevant caveats about your analyses as well as logical next steps (concerning data, models, and inferences) in the back end.

29. Some of you did not provide histograms to assess model fit. Visualizations are very important not only for writing a good paper but also for evaluating model fit and drawing insight from the model/data.

30. Make sure you proofread, spellcheck, etc. Take more pride in your work! It's advisable to double space and leave adequate margins for comments.

31. Be careful with the word "significant." It has a narrow technical meaning when we're dealing with these models and therefore it's best avoid using it in other contexts within your paper.

32. Ditto for "random." All of our models are random. It's just that we're better able to characterize some patterns better than others. That doesn't make them any less random.

33. Avoid too much "MBA speak." This is a relatively narrow, technical assignment, so don't make overreaching claims that aren't tightly tied to the models you examined.

34. Avoid unnecessary precision! It's bad enough to include cut-and-paste output with 7-8 significant digits in it, but I was really dismayed at how many of you reported those overly precise numbers in text as well.

Please reread this set of guidelines as you go through the process, and keep it handy for the subsequent project as well…